2018 Ph H2 Q5

Section: Our Dynamic Universe

Topic: Evidence for the Big Bang & the Hubble constant

Hubble’s Law states the Universe is expanding. The expanding Universe supports the Big Bang theory.

(a) State one other piece of evidence supporting the Big Bang.
(b) A student uses Hubble’s 1929 data and draws a line with gradient $H_0 = 2.0\times10^{-17}\ s^{-1}$.
    (i) Calculate the age of the Universe using age = $1/H_0$, in years.
    (ii) The current estimate is 1.8×10^{9} years. (A) Explain the difference. (B) Suggest how the student could improve their graph to get closer to the current estimate.
(c) State what physicists think causes the observed increase in the rate of expansion.

Worked solution

(a) Any one of:

• Cosmic Microwave Background (CMB) radiation (blackbody at ≈2.7 K).
• Abundance of hydrogen and helium / light elements.
• (Resolution of) Olbers’ Paradox (night sky is dark).

(b)(i) Age from H_0

age = $1/H_0 = 1/(2.0\times10^{-17}\ s^{-1}) = 5.0\times10^{16}$ s.
Convert to years: $5.0\times10^{16} / (3.156\times10^{7}) \approx 1.59e+09$ years.

Answer: $\approx 1.6\times10^{9}$ years

(b)(ii)(A) The student's H_0 is too large / not accurate; the line of best fit is incorrect; or it uses limited/older data. More (especially distant) and more accurate data give a smaller H_0 and hence a larger age estimate.

(B) Draw a correct line of best fit and/or include a larger sample — ideally current data — particularly at large distances.

(c) Dark energy (drives the accelerated expansion).

Final answers

(a) e.g. CMB at 2.7 K / light-element abundances / Olbers' paradox.

(b)(i) age ≈ 1.6×10^{9} years.

(b)(ii)(A) Student's H_0/fit/data is inaccurate; (B) use better line of best fit and more/current data.

(c) Dark energy.

Revision tips

- From Hubble's law, the age of the Universe is $\approx 1/H_0$.

- Older/sparser datasets give different H_0 estimates; modern fits use many distant galaxies.
- CMB and primordial abundances are key Big Bang evidence beyond expansion alone.
- Accelerating expansion is attributed to dark energy.